

Assessing instream habitat suitability and hydraulic signatures of geomorphic units in a reconstructed single thread meandering channel

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ABSTRACT

Stream channel reaches consist of a series of geomorphic units (e.g. pools, runs and riffles) that mitigate flow velocity and provide a variety of aquatic habitat. The hydraulic signature of each geomorphic unit is unique and should remain so during variable flow conditions in order to satisfy habitat requirements. We use the hydraulic signatures measured at different geomorphic units in a recent channel realignment project to evaluate suitability of reaches and geomorphological units for target fish species. In 2009, 650 m of creek and valley were realigned in Brampton, Ontario, to facilitate development. In 2013, water velocity, depth, temperature and channel substrate were measured at cross-sections between 1 May and 15 October. The assessment illustrated that while geomorphic units have unique signatures during moderate to low-flow conditions, at high-flow conditions the boundaries between geomorphic units blurred. Furthermore, our findings suggest that suitable water depth, velocity and temperature conditions existed at least 68% of the time over the entire study period but was reduced during the estimated spawning period for the target species. Specifically, water depth was only ideal for approximately 38% of the spawning period. Finally, in cases where both geomorphic and ecologic systems interface (e.g. channel restoration), it is important to view the whole picture. This study contributes to the on-going discussion on restoration and success and is particularly useful in light of recent stream restoration trends in rapidly expanding urban regions around the globe. Copyright © 2015 John Wiley & Sons, Ltd.

KEY WORDS stream hydromorphology; restoration; aquatic habitat; hydraulic variables; species-at-risk

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INTRODUCTION

Human activities within a watershed, such as agriculture, forestry and urbanization, impair or alter aquatic ecosystems (Maddock, 1999; Moerke *et al.*, 2004;). This impairment is especially evident when streams and channels are moved or modified to utilize the land surface around that stream (e.g. relocating streams to the edges of agricultural fields, burying streams for parking lots or road networks). Channels that were once abundant in fish species now have reduced populations (McKee and Parker, 1982; Parker *et al.*, 1988; Andersen, 2002; Poos *et al.*, 2012). Stream restoration projects aim to undo damage and in many cases seek to improve aquatic habitat where possible by ensuring suitable spawning, foraging and other life cycle habitat requirements (Newson and Large, 2006; Palmer and Bernhardt, 2006; Carré *et al.*, 2007; Brierley *et al.*, 2010; Salant *et al.*, 2012; Gostner *et al.*, 2013a,

2013b; Marttila *et al.*, 2015). There are several studies that evaluate geomorphic units, biotopes and available habitat in natural channels (e.g. Newson and Newson, 2000; Clarke *et al.*, 2003; Lamouroux and Jowett, 2005; Moir and Pasternack, 2008), but there is limited work focused on the hydraulic signatures of specific design elements as it relates to habitat availability in restored streams beyond engineering works (e.g. Gostner *et al.*, 2013a, 2013b).

Measuring physical features on both reach and the geomorphic unit scale are used to characterize the variable driving and resisting forces within a stream channel and assess the habitat available in a channel (Padmore, 1998; Newson and Newson, 2000; Thomson *et al.*, 2001; Newson and Large, 2006; Palmer and Bernhardt, 2006; Milan *et al.*, 2010). Habitat suitability is inherently linked to geomorphic units and biotopes, by evaluating the hydraulic signatures of geomorphic units through various flow conditions we can begin to develop methods to assess stream morphological resilience and the availability and quality of available habitat in a given stream restoration project. Geomorphic units have a distinct depth, flow and substrate, which when nested within a stream reach,

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provide a variety of flow features and therefore potential habitat (Hawkins *et al.*, 1993; Dollar, 2000; Newson and Newson, 2000; Church, 2002; Clarke *et al.*, 2003; Gostner *et al.*, 2013b). Finally, there have been notable advances in the way hydraulic signatures for instream habitat is characterized (e.g. Marcus *et al.*, 2003; Milan *et al.*, 2010).

Jowett (1993) compared the physical conditions at a location to a subjective classification of habitat type (geomorphic unit) at that location and determined that riffles and pools were easily distinguishable but transitional geomorphic units (generally termed runs) were relatively more challenging to distinguish from riffles. These results have implications for stream restoration and channel design in that the functions that each geomorphic unit provides are distinct and therefore provide different habitat functions. The original work by Jowett (1993) focussed on low-flow conditions, and work by Padmore (1998) further investigated hydraulic signatures across a range of flow conditions to demonstrate water level was an important factor for several hydraulic biotopes. Additionally, Froude numbers and velocity-depth ratios were considered valuable determinants between geomorphic units (Lamoureux and Jowett, 2005; Moir and Pasternack, 2008).

Restored streams can improve habitat availability through design elements with varied morphological, sedimentological and hydraulic characteristics. In this study, we attempt to bridge what Jowett and others have described as the functional hydraulic signatures of geomorphic units to what is known about stream restoration ecology (for a recent analysis see Marttila *et al.*, 2015). Throughout the late spring until fall 2013, we monitored a recent corridor realignment, stream and habitat restoration project in the Greater Toronto Area (GTA) near Brampton, ON, Canada. Our working hypothesis was that sufficient geomorphic unit construction ensured that the restoration project was able to provide suitable habitat for species-at-risk. Furthermore, evaluating the hydraulic signatures of geomorphic units illustrate that subjective evaluation of stream restoration projects does not capture the complexity and flexibility (and therefore potential resilience) necessary for stream and habitat restoration successes. This work contributes to our overall understanding of geomorphic units as important, yet understudied elements within restored streams. Finally, given the landuse/land change pressures associated with development around urban centres like the GTA, this study serves as a useful case study to evaluate the impact of integrating stream realignment projects with stream restoration projects and overall habitat improvement.

METHODS

Study site

Sections of stream valley (Tributary 8b; ~ 3.5 km²) within the Credit River watershed (795 km² total area), in

Brampton, Ontario were realigned and restored to facilitate development of previously agricultural land into residential and light commercial land use (Figure 1). Tributary 8b is part of the lower Credit River watershed where landuse is highly urbanized, and newly developed stormwater systems are integrated with stream channel restoration efforts. To meet permitting requirements under the Endangered Species Act in Ontario Canada, the channel realignment needed to provide suitable habitat for reddsides (*Clinostomus elongates*), a designated minnow species-at-risk [Committee on the Status of Endangered Species in Canada (COSEWIC), 2007]. As such, this project was one of the first in the GTA where ecological system service function was integrated into the management of surface water for reddsides (Geomorphic Solutions, 2009). Since construction completion in 2010, annual monitoring consisted of one longitudinal profile and six cross-sectional surveys usually completed in June (Geomorphic Solutions, 2009, 2011, 2012, 2013; GHD, 2014).

Both the Ministry of Natural Resources and Credit Valley Conservation identified Tributary 8b as reddsides survival habitat; therefore, permission was required under Section 17(2) of the Endangered Species Act [Committee on the Status of Endangered Species in Canada (COSEWIC), 2007], even though reddsides were not found in the reach prior to restoration (Geomorphic Solutions, 2009). The restored stream channel and valley alignment were designed to provide reddsides habitat, in effort to promote reddsides recovery and their potential return to the reach (Geomorphic Solutions, 2009). Design elements that provided habitat for reddsides include vegetation plantings that would overhang channel banks to provide shading, undercut banks and instream cover such as woody debris and boulders. These features are typical design elements in use elsewhere. Because reddsides are commonly observed with creek chub (*Semotilus atromaculatus*) and reddsides use creek chub nests for spawning, the habitat requirements of the two species are similar (Koster, 1939; McKee and Parker, 1982; Novinger and Coon, 2000; Andersen, 2002; Zimmerman, 2009). Although specific substrate requirements are not well documented, the impacts of siltation on minnow species is well-known to negatively impact feeding and spawning (e.g. Berkman and Rabeni, 1987).

Stream flow around the GTA is a function of snowmelt and rainfall; during this, study runoff was generated by rainfall between May and October (Environment Canada, 2014). Where development or relict agriculture has not modified the surface, sand and gravel deposits surrounded by Halton till make-up the surficial material in the area. The restored Tributary 8b includes a corridor width of 50 to 70 m, with riffle-pool sequences, a meander planform and continuous channel connection

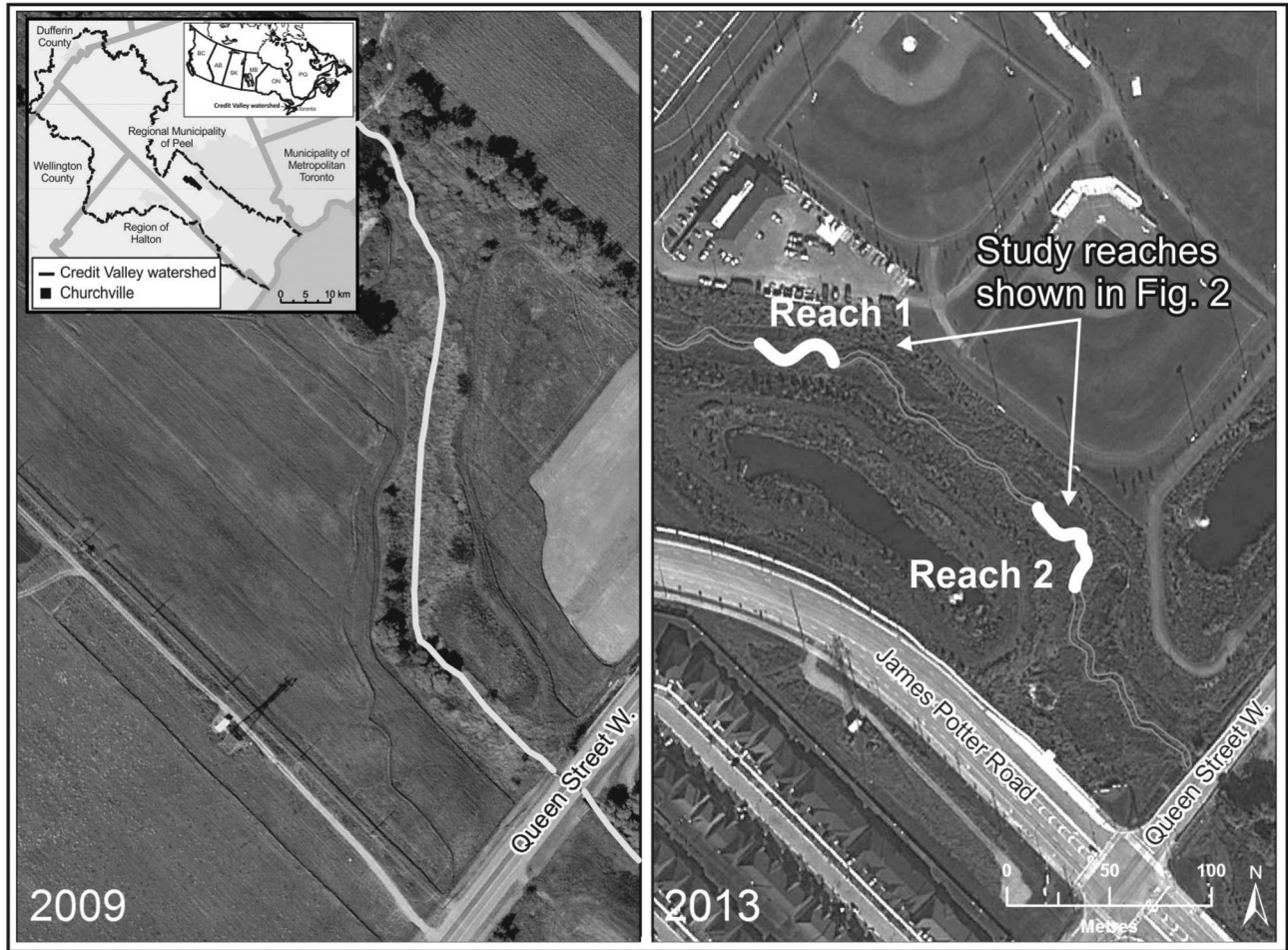


Figure 1. Realigned stream channel and study reaches. The inset map shows the study location within the Credit Valley watershed in the northwest region of the Greater Toronto Area, Ontario, Canada. The photos show before (2009) and after (2013) construction of the stream channel alignments in the study area (the scale in each photo is the same and is shown in the bottom right of the 2013 photo).

between upstream and downstream. Other design features include pocket wetlands, native riparian vegetation, and rootwads and wattles for bank stabilization and overhanging cover for fish (Geomorphic Solutions, 2009). The riffle-pool sequences and bioengineering elements are to enhance aquatic habitat with special consideration for reddsides dace.

Field methods

The restored channel was monitored and surveyed several times between 15 May 2013 and 15 October 2013. Initial site visits in early spring were used to identify and classify study cross-sections within the study reaches (Figure 2). The restored channel was subjectively classified (e.g. pool, run and riffle) based on the water surface conditions observed during the initial survey and follows the procedures outlined in Jowett (1993). It should be noted that the simple pool, run and riffle designations are a

simplification and were used to align with post-construction survey data. For further reading on biotopes and other habitat types, please refer to Newson and Newson (2000). Pools had slow-flow and a smooth-water surface, riffles had swiftly flowing water with a high proportion of the surface broken, and runs were between pools and riffles with a somewhat undulating surface (Figures 1 and 2). Each cross-section was perpendicular to bankfull flow and marked with a pin on each bank. The study area was divided into two reaches, the upper reach (Reach 1) had 18 cross-sections (#22–39), and the lower reach (Reach 2) had 21 cross-sections (#1–21)(Figure 2). Water velocity and depth measurements were collected at each cross-section using an Acoustic Doppler Velocimeter at a variety of flow conditions, over 12 visits. Measurements were collected within the bankfull channel at 0.1 m intervals and at 60% of the water depth with minimal disturbance to the adjacent riparian area and cover. In addition, a Heron water level logger measured and

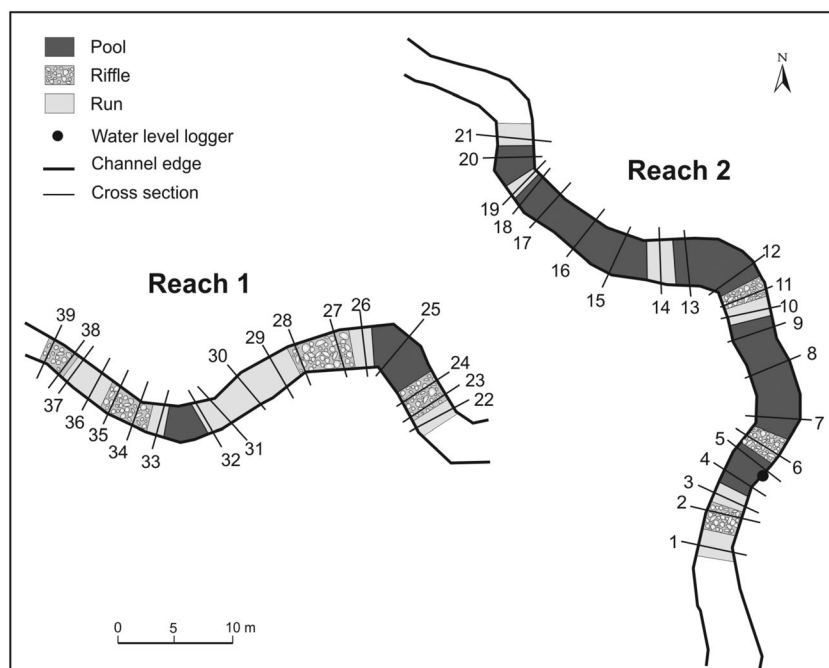


Figure 2. Two reaches with various geomorphic unit sequences were the focus of velocity-depth profiles, channel mapping and analysis. Each cross-section is numbered on the figure; water level and water temperature was monitored between cross-sections 4 and 5 on Reach 2 (dot).

recorded absolute pressure (water and atmospheric pressure ($\pm 1\%$) and water temperature ($\pm 0.2^\circ\text{C}$) at 10-min intervals within a cross-section in Reach 1 (Figure 2). A Heron Barometric logger ($\pm 1\%$) measured and recorded atmospheric pressure near the water level logger but not in the water. Daily rainfall was recorded and provided by Credit Valley Conservation at a site $\sim 15\text{ km}$ north of Tributary 8b.

Topographic survey of the site and each cross-section was completed using a Leica real-time kinetic global positioning system (accuracy from 0.01 to 0.02 m). Inflexion points at each cross-section were captured and continued to the bank at approximately 0.1 m intervals. The surveyed cross-sections continued to at least 2 m beyond the top of banks to capture the floodplain where the interval between points increased to 0.2 m. Survey methods were similar to the survey method used for the initial as-built survey and annual monitoring at Tributary 8b (Geomorph Solutions, 2009, 2011, 2012, 2013; GHD, 2014). Additional notes on rootwad delineation (21 Sept), the channel centreline (18 June, 21 July and 14 and 21 Sept), and within the top of banks, floodplain and valley top and toe (21 July) were surveyed.

Channel substrate was measured using two methods: quadrat and pebble counts. After water velocity and depth measurements were taken, a quadrat was placed on the channel bed at a cross-section. The quadrat method was used in order to minimize bed disturbance. Photographs were taken with a waterproof camera, and 40 particles were selected from in the photo, measured and tallied to

determine the median grain size at each cross-section. A modified-pebble (particles $> 2\text{ mm}$) count method was used to determine the surface substrate size distribution in September (Wolman, 1954; Potyondy and Hardy, 1994; Kondolf, 1997). Particles ($> 2\text{ mm}$) at each cross-section were measured along the intermediate axis; only 40 particles were measured at each cross-section given the narrowness of the channel.

RESULTS

Geomorphic changes since 2010 construction

In order to evaluate design features, it was important to account for changes or variability in the restored stream since the as-built survey. There were minimal differences in the primary characteristics reported in the as-built survey completed in 2010, annual review reports completed each year since 2010 and the surveys conducted in this study (Table I). The as-built survey and annual reviews do not evaluate runs, thus, these features were not included in the annual comparisons. The differences because construction have been minimal, with the exception of bankfull width at riffles and pools, which appear to be widening (Table I). While the Froude (Fr) number for riffles and pools remained less than one (sub-critical), maximum shear stress, determined from the average gradient and depth for each geomorphic unit observed at bankfull conditions on 9 July 2013, decreased slightly (less than 10% change) in both geomorphic unit types across the site (Table II).

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Table I. As-built 2010, annual review from 2011 to 2013 (GHD, 2014), and 2013 study measurements.

Characteristics	2010 ^a	2011	2012	2013	This study
Field survey date (mm/dd)	01/20 ^a	06/15	06/7–8	06/18	See Fig 3
Channel bed gradient (%)	0.55	0.48	0.54	0.54	0.54
Riffle bed gradient (%)	1.58	3.10	1.74	1.70	1.46
Riffle length (m)	5.00	5.60	5.70	5.70	5.68
Riffle-pool spacing (m)	14.20	13.20	10.60	11.80	10.80
Bankfull width riffles (m)	2.75	2.71	3.00	3.22	3.40
Bankfull depth riffles (m)	0.41	0.38	0.37	0.44	0.39
Bankfull width pools (m)	2.81	2.70	2.90	3.04	3.80
Bankfull depth pools (m)	0.61	0.47	0.49	0.51	0.60
Grain size (mm) D ₅₀	n/a	17	n/a	14	Riffle: 12, Pool: 1 (See Fig 5)

^a Construction completed in November 2010, survey completed in January 2011.

Table II. Froude number and shear stress in 2010 and 2013 for riffles and pools.

	2010		2013	
	Riffle	Pool	Riffle	Pool
Max. shear stress (N/m ²)	23.94	35.90	20.03	30.59
Froude number	Sub-critical		Sub-critical	

Hydraulic signatures of geomorphic units through variable summer flow

Between May and October 2013, the field site was visited on 12 days to survey water depth, velocity and observe flow conditions; on four visits, substrate was measured (Figure 3). Continuous water level and temperature measurements were recorded and used to contextualize survey day conditions. High-water levels were observed on 11 June and 9 July. The largest rain event (~52 mm, July 8) produced significant flow conditions and a site visit on July 9 indicated that the banks were overtopped as evidenced by a heavily saturated floodplain and in some areas water draining from the floodplain back into the channel as overland flow. The 11 June event was the third largest (~31 mm over three days), and conditions were near bankfull. Observations on 29 May, 11 June and 9 July represent high-flow events and observations taken on 17 May, 19 June 4 and October represent low-flow events.

Water depth and velocity measurements were recorded at 0.1 m intervals at individual cross-sections and compared to the subjective geomorphic unit classification (e.g. riffles, pools, runs) similar to Jowett (1993) to evaluate hydraulic signatures in different habitat types through different flow conditions (Figure 4). Pools were deep with lower velocities; riffles were shallow with higher velocities, and runs fit between riffles and pools. During higher-flow events, water depth-velocity distribution for riffles at high-flow events were such that they took on the hydraulic

properties typical of runs in moderate to high-flow events (Figure 4), which corresponds to observations during field work. We did not convert our stage data to flow because the channel and watershed was small (~3.5 km²). Flow data at this site prior to construction did not exist as this channel was essentially a ditch at the edge of a field (Figure 2), and flow was not monitored.

Substrate size at each cross-section fluctuated through the season, most notably in typically deeper features (Figure 5). Although particle size distributions in riffles, pools and runs were distinctively different from each other throughout the season, there were small changes in pools and runs. At various times throughout the season, the substrate distribution in pools was well graded and coarsened slightly through the summer (Figure 5a). The particle distributions in the run sub-reaches sampled were well graded to uniformly graded and only coarsened slightly when considering the change since the May observations (Figure 5b). In riffles, the particle distribution was uniformly graded and arguably showed some signs of gap grading (particles sized between ~20 mm and 50 mm were absent in the 21 July survey), but the change in distribution was the smallest observed of the three geomorphic unit types (Figure 5c).

Habitat availability

Overall, habitat availability is considered relative to life cycle activities as reported in the literature for species

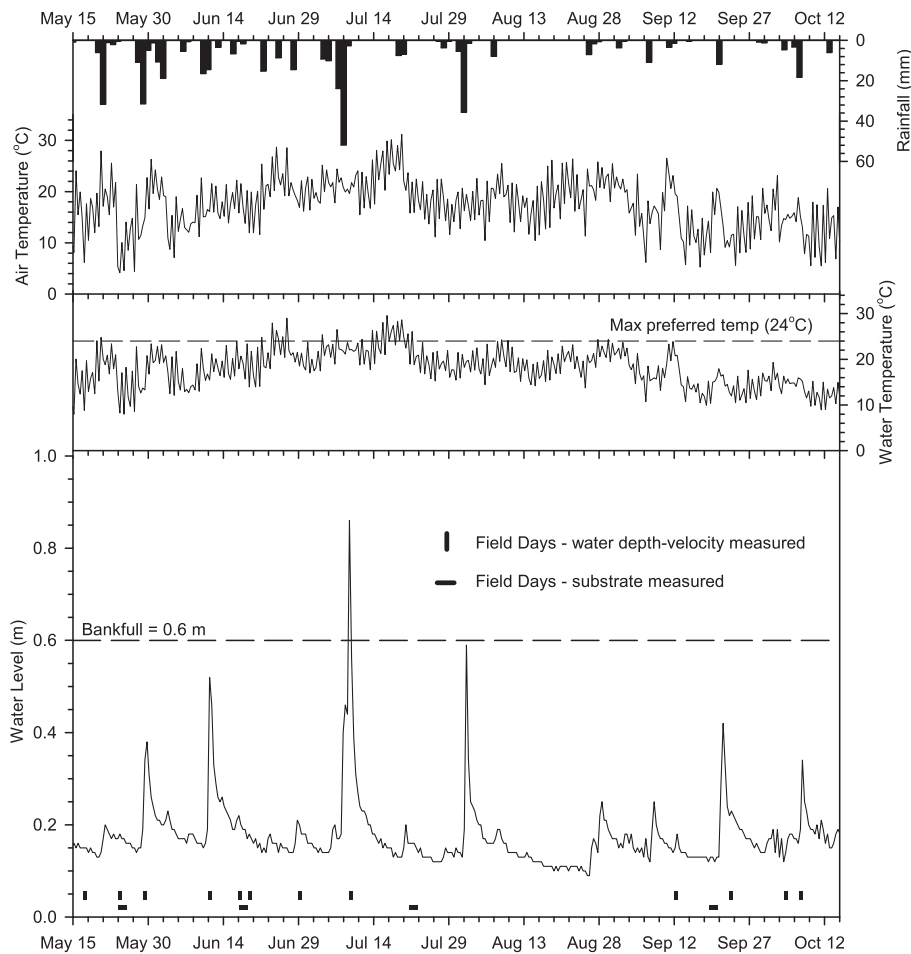


Figure 3. Rainfall, air temperature, water temperature, water level and field days. Maximum preferred water temperature is shown (24 °C: Novinger and Coon, 2000). Bankfull level at the water level logger is 0.6 m.

commonly observed with redbreasted sunfish (e.g. creek chub; Koster, 1939; McKee and Parker, 1982; Novinger and Coon, 2000; Andersen, 2002; Zimmerman, 2009). Evaluation of potential habitat for redbreasted sunfish was determined as a proportion of measurements that fall within the ideal range and included water depth ($n=3243$), velocity ($n=3220$) and temperature ($n=31454$) through the study period (Figure 6). Water depth and velocity were within the ideal range 86% and 82% of time, respectively, while water temperature was only within the ideal range for 66% of the measurements (Figure 6).

Spawning season occurs in the last 2 weeks of May in riffles (Koster, 1939). Spawning habitat evaluation included only the water depth ($n=226$) and velocity ($n=215$) measurements completed on 17, 24 and 29 May, in riffles (Figure 6). Because the water level logger was located in a pool, the water temperature is reflective of that of the pool and is not necessarily reflective of conditions in riffles ($n=2248$). Spawning habitat is assessed separately from overall habitat because spawning habitat assesses egg survival, and overall habitat assesses adult redbreasted sunfish

survival. Although, ideal spawning habitat conditions existed, the proportion of measurements was much lower compared with the overall habitat conditions (Figure 6).

Formal fish surveys were not conducted as part of this study, but observations while in the field and during site visits noted fish presence. Perhaps, most significant is that creek chub was commonly observed throughout the field season. Redbreasted sunfish typically coexist with creek chub as the creek chub nests are used for redbreasted sunfish spawning (Koster, 1939; Scott and Crossman, 1973). Other minnow species were observed during the field season including western blacknose dace (*Rhinichthys obtusus*).

DISCUSSION

Flexible resilience in restored channel designs

A natural channel has deformable boundaries, such as channel banks and beds, which change overtime due to natural processes, such as high-flow events (Gillilan *et al.*, 2005; Palmer *et al.*, 2005; Wohl *et al.*, 2005). Deformable

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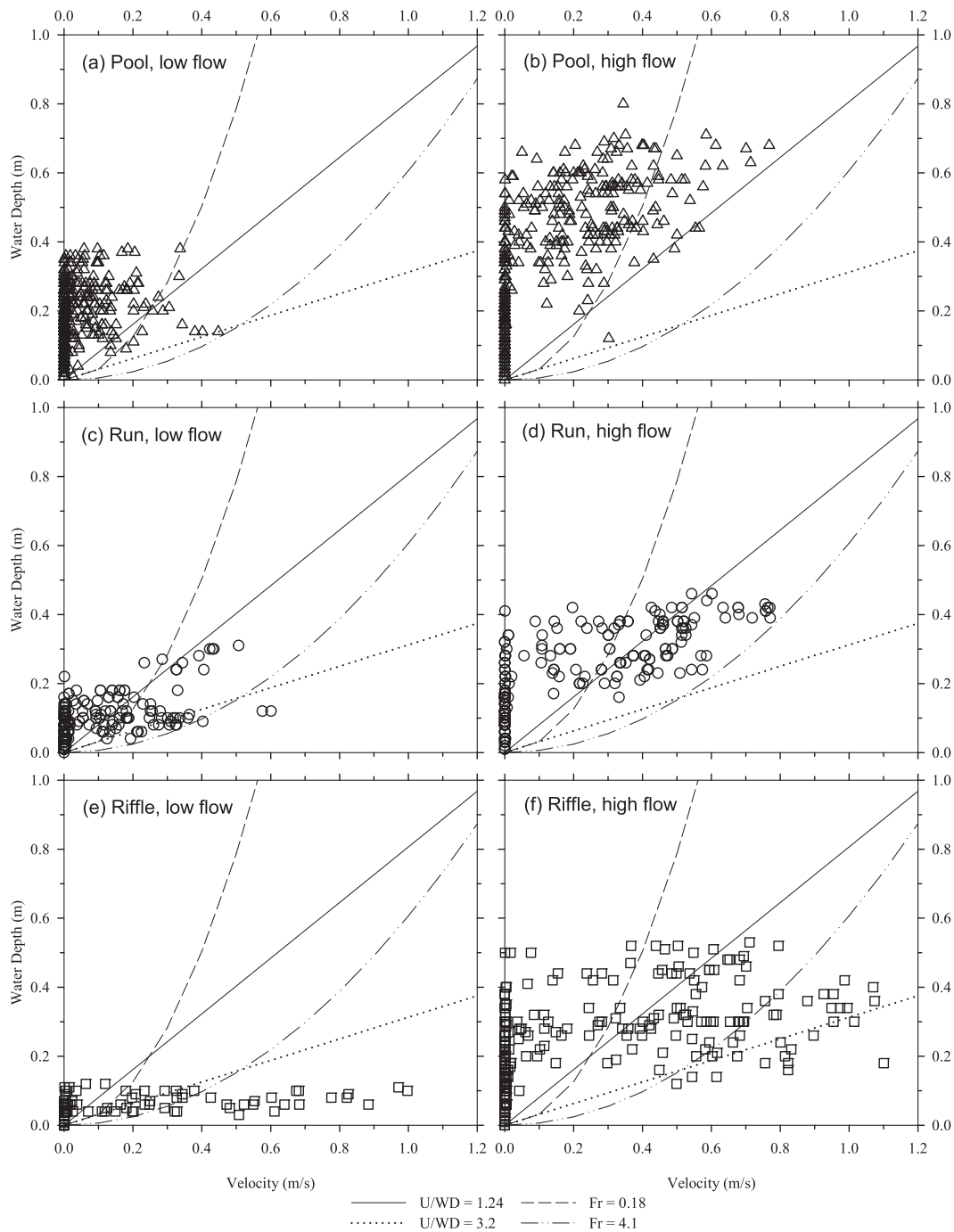


Figure 4. Water depth-velocity plots through different geomorphic units at low (a, c and e) and high (b, d and f) flows. The velocity/water depth ratio and Fr numbers are given as the typical hydraulic signature limits for pools and riffles. Pools (a and b), transitional runs (c and d) and riffles (e and f) are plotted separately to show the unique hydraulic signatures; these geomorphic units display at this site during low-flow. These signatures seem to blur during high-flow conditions.

boundaries are important design characteristic in channel restoration projects but can be an added challenge when restoration is confined to a corridor or required to interface with existing infrastructure (e.g. road networks, stormwater systems and property owners). The restored Tributary 8b reach showed minimal changes in 2013 since the 2010 as-

built survey (Table I). However, there were changes within channel geomorphic units such as riffle-pool spacing and bankfull widths, suggesting that in some ways these constructed geomorphic unit boundaries were 'blurring' in response to forcing conditions (e.g. variable flow). That the hydraulic conditions remained similar (Table II) and

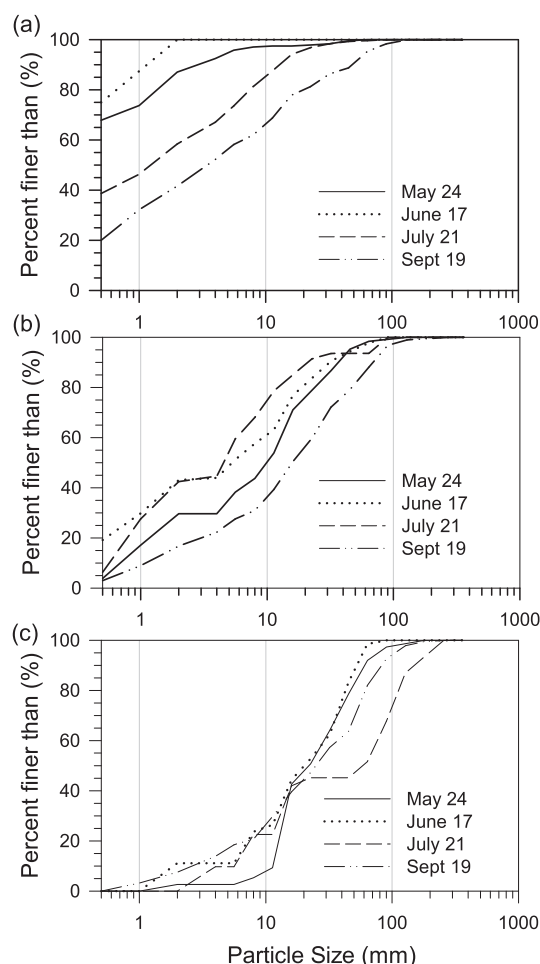


Figure 5. Channel substrate cumulative distribution for (a) pools, (b) runs and (c) riffles for each day measured.

that specific design features were still in place (e.g. riffles and pools were identifiable) following high-flow conditions, suggests that the overall design was flexibly resilient. Even with rain and snowmelt events, no repairs to the restored reach were needed since the 2010 construction. Several studies report that a resilient and self-sustaining restoration project does not require repair after flow events (e.g. Palmer *et al.*, 2005; Lake *et al.*, 2007).

During variable flow conditions, pools, riffles and the transient geomorphic unit runs, performed as designed, ensuring that habitat heterogeneity was preserved (Gostner *et al.*, 2013a). In addition, to minimal channel adjustments since the 2010 as-built survey, monitored cross-sections of various geomorphic units show evidence of hydraulic signature blurring (Figure 4). As velocity and water depth increased, the differences between geomorphic units decreased. For the most part, pools maintained their hydraulic signatures, with most observations coincident with velocity-water depth ratios less than 1.24 and a Froude number less than 0.18 (Figure 4a, b). Riffle geomorphic features demonstrated the most variance between low and

high-flow conditions (Figure 4e, f). Riffles were described as ‘drowned’ during high-flows (Booker *et al.*, 2001; Emery *et al.*, 2003). During these high-flows, there were significantly fewer observations coincident with lower limits of typical riffle hydraulic characteristics (velocity-water depth ratios greater than 3.2 and Froude number greater than 4.1). The reduction or decreasing distinction between geomorphic units during high-flows could have consequences for ensuring or maintaining habitat heterogeneity necessary for specific species activity (e.g. higher-flows required for spawning). Emery *et al.* (2003); measured velocity at different geomorphic units at low and high-flows and found geomorphic units were indistinguishable with an increase in water depth. The restored reach in this study was behaving like a natural channel (Booker *et al.*, 2001; Emery *et al.*, 2003).

Despite the potential for higher-flow events to alter geomorphic units and larger channel reach characteristics, persistent patterns in channel substrate suggest that the stream is in a balanced or near-balanced state (Figure 5). Flexible resilience within the channel design is critical to achieving this balance and thus maintaining created habitat availability. Substrate observations throughout the study period illustrated minor changes in the bed (Figure 5). Most important, although is the preserved particle size distribution for each geomorphic unit type. The slight coarsening in pools over the field season is an important process and likely reflects accumulation of finer-grained material during lower flows, rather than significant scour (Keller, 1971; Lisle, 1979; Kondolf and Wilcox, 1996; Milan *et al.*, 2001). Furthermore, several studies recognize that changes following high-flow events are to be expected (Gillilan *et al.*, 2005; Palmer *et al.*, 2005, 2007).

‘If you build it, they will come’ – Stream Restoration and Habitat Heterogeneity.

Stream restoration projects that aim to provide some ecosystem service such as suitable habitat for target species, often subscribes to the ‘field of dreams’ principle, ‘if you build it, they will come’. Founded in ecological theory, this principle links habitat heterogeneity with species diversity, such that if a restoration project builds habitat heterogeneity, increased species diversity should follow (e.g. Palmer *et al.*, 1997, 2007, 2010). Marttila *et al.* (2015) evaluated the long-term (>10 years) habitat heterogeneity and species diversity on restored low-order Boreal forested streams in Finland. Their findings suggest that although the restored channels maintained restored features (e.g. habitat heterogeneity), species diversity did not improve. In this study, similar findings were made over a shorter period. Although, the restored channel features showed minimal changes to the as-built survey (Table I and II), and design elements were included to specifically

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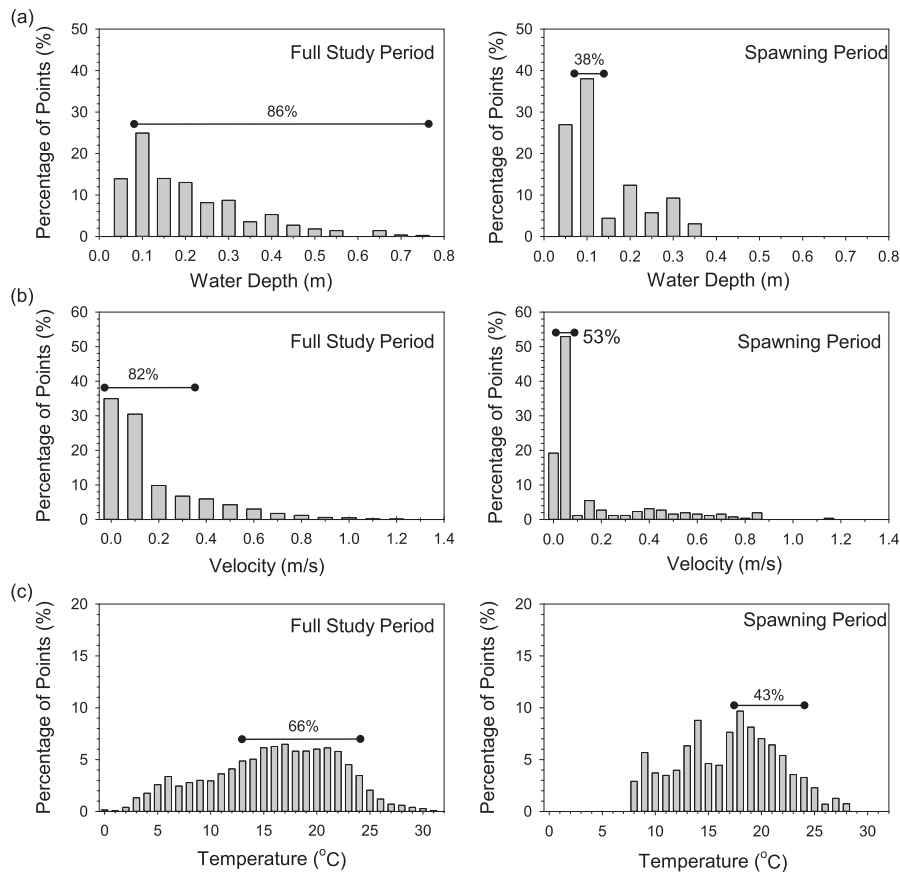


Figure 6. Habitat suitability based on (a) water depth, (b) velocity and (c) water temperature through the field season (left side) and the spawning period (right side).

address reddsides habitat requirements (species-at-risk), reddsides populations were not observed. Reasons for no reddsides populations might include, lack of habitat connectivity or disturbance elsewhere (e.g. on-going construction and restoration, both up and down stream of the field site), lack of overwintering habitat (e.g., Davis *et al.*, 2015), and potentially degraded water quality (e.g. urban runoff, high turbidity due to construction; and water temperature).

Novinger and Coon (2000) completed a reddsides dace study that observed maximum water temperatures of 30 °C over a 2 week period and found no mortalities. On 17 July 2013, over a 4-h period, water temperatures over 30 °C were recorded at the logger site (Figure 3). The water level logger water temperature is sheltered and of a median depth in a pool; it is reasonable to suggest that water temperature would have been warmer in shallower sections of the channel, but it was important to measure this variable in a place where fish were likely to seek refuge in times of stress. From the Novinger and Coon (2000) study, a maximum preferred temperature was determined as 24 °C. During the 2013 study period, 5% of the water temperature measurements in the restored reach exceeded 24 °C

(Figure 6). Should reddsides dace been observed, it is possible that high temperatures in areas of refuge would have caused stress to the species-at-risk.

Species require specific water depth, velocity and temperature preferences to carry out their life functions such as foraging, resting and spawning (Koster, 1939; McKee and Parker, 1982; Novinger and Coon, 2000; Zimmerman, 2009; Poos *et al.*, 2012). Within Tributary 8b, the Ministry of Natural Resources identified reddsides dace as the species-at-risk (Geomorphic Solutions, 2009). The Tributary 8b design brief states that reddsides dace was not found within the immediate area, but habitat requirements were considered for the restored reach (Geomorphic Solutions, 2009). However, reddsides dace coexist with creek chub (*Semotilus atromaculatus*) and use creek chub nests for spawning (Koster, 1939; Scott and Crossman, 1973). Creek chub were commonly observed throughout the study period.

Spawning habitat provided by the restored Tributary 8b reach is lower compared to the overall habitat (Figure 6). Reddsides dace can seek shelter at the rootwads from high velocities (He *et al.*, 2009; Zimmerman, 2009). Unfortunately, once laid, eggs cannot survive high velocity

conditions by seeking refuge. Eggs are non-adhesive and sensitive to high velocity (Koster, 1939; Scott and Crossman, 1973). Such conditions occurred on 29 May, which could have swept the eggs away (Figure 3).

Koster (1939) found the lowest temperature, reddsides were observed, and spawning was at 18 °C. Approximately, 52% of the water temperature measurements were less than 18 °C (Figure 6). This was due to diurnal water temperature fluctuations from 15 to 29 May (Figure 3). The Koster (1939) temperature limit estimates were based on daytime field observations when reddsides were visible and coincidentally when the water was warmer; consequently, these limits may be conservative.

Research on reddsides habitat is somewhat limited and much of what is known is based on habitat requirements for creek chub as the two species are found together (Novinger and Coon, 2000; Zimmerman, 2009). Work on reddsides habitat, in particular, overwintering habitat is on-going (e.g. Davis *et al.*, 2015) and will be crucial to understand what is needed for their survival and how channel restoration works might incorporate habitat features into designs. However, more information may not address the larger 'field of dreams' problem that seems to be occurring at this site and is noted elsewhere (e.g. Boreal forest stream in Finland, Marttila *et al.*, 2015). Although legislation in Ontario, Canada, requires species-at-risk habitat availability be included in designs, it may not be successful for (re)establishing populations in these restored channels. Certainly, landuse/land change pressures on streams within urban and expanding metropolitan areas require that stream restoration projects be mindful of species-at-risk needs, but perhaps a view beyond a specific target species would be more beneficial.

CONCLUSION

In a recently completed channel alignment and stream restoration project, we evaluated the short-term changes in the channel design and the hydraulic signatures of geomorphic units to assess habitat heterogeneity and therefore availability for a designated species-at-risk. Given design elements that are flexibly resilient, habitat heterogeneity has persisted since construction. Moreover, as water levels and velocities change through geomorphic units, distinctive hydraulic signatures blur, most notably riffle habitat decreases. Understanding how stream restoration projects have fared because construction is essential for functional designs and the overall success.

Finally, although restoration designs and intent was to provide suitable habitat for a minnow species-at-risk, populations of this species have not established in this channel. Further studies focused on habitat needs for the species-at-risk, and a larger survey to assess potential limits

to population establishment (e.g. overwinter habitat, water quality and habitat connectivity) could further improve restoration efforts. This study presents a useful case study into the impacts and successes of a large-scale channel realignment and restoration project in a rapidly transitioning urban region.

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